Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/11/2005
PI Name:	Bernholc, Jerry Ph.D.		
Project Title:	Growth and Properties of Carbon Nanotubes		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-8202	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-02
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roland, C. (North Carolina State University)		
Grant/Contract No.:	NAG8-1479		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		
Task Description:	Using both classical and quantum molecular dynamics simulations, we are in the process of investigating the growth, mechanical and electronic properties of nanotubes, as well as their usefulness as a new material for energy storage. We have investigated ferroelectric polymers and predicted new polymers with much improved properties. Nanotubes are perhaps one the most exciting materials to have emerged this decade. The interest and excitement derives from their unusual structural and electronic properties. Experiments and theory indicate that they are super-strong fibers with unique electronic properties. These properties should lend themselves to the development of unique applications as composites, catalysts, field emitters, photonic materials, molecular straws, molecular devices and as MEMS materials. Our objective is to theoretically study the growth and properties of these materials, and to propose new advanced materials based on theoretical predictions. We have also predicted new ferroelectric polymers. In the field of nanotubes,		

	theory has played a significant role in elucidating some of the very exciting possibilities that these unique materials offer. Indeed, theoretical predictions of carbon and BN nanotube formation predate their experimental discovery. Other important milestones include theoretical predictions of their mechanical and electronic properties. Theoretical studies of nanotube materials and their properties have therefore played a particularly important role in this field. Our prediction of novel ferroelectric polymers will likely have impact on the design of new actuators, sensors and switches.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The predicted novel materials, such as new nanoscale gas sensors and new ferroelectric materials, may have a significant and wide-ranging impact on technology.
Task Progress:	Prediction of new superpolar polymers Based on our prior work on polarization and piezoelectric properties of boron nitride nanotubes, we investigated the properties of the most important ferroelectric polymers, namely those in the PVDF family. Our calculations used state-of-the-art first principles methods, such as the Berry phase formalism and recently also Wannier function construction and properties. The results reproduce very well the observed ferroelectric and piezoelectric constants of these polymers, as well as of their co-polymers. We have analyzed our results in terms of charge transfer, dipole formation and the direction of polarization, using Wannier functions and decomposition into monomer contributions. From our results, we realized that a substantial increase in polarization properties could be expected from the substitution of the carbon polymer backbone with BN units. Indeed, after the substitution our first-principles calculations find a very substantial increase in the polarization properties. The best polymer offers about a 100% improvement in the piezoelectric response and thermal stability over the commonly used polyvinylidene fluoride (PVDF). Furthermore, the methods for syntheses of these new polymers are readily available. In fact, some of those have already been used as precursors in synthesis of other materials, such as BN nanotubes, but they have not been investigated as ferroelectric materials in their own right. The above polymers could find many uses in advanced applications, such as sonar sensors, nanoactuators and switches, flexible airplane rings etc. To our knowledge, our prediction constitutes the first case where an improved ferroelectric was "designed" on a supercomputer. Based on the results of these investigations, we have initiated collaboration with experimentalists aimed at the development of advanced polymers with enhanced ferroelectric and deposition properties. This work has resulted in the discovery of a new, high-temperature ferroelectric polymer. Hydrogen Storage in Li-doped Carbon Nanotube Systems Issues related to hydrogen storage in carbon nanotube systems are of great interest because of high potential for technological applications. Specifically, initial experimental investigations of hydrogen uptake in Li-doped carbon nanotube systems were particularly promising. We have investigated this issue theoretically in both graphite and carbon nanotube systems with large-scale density calculations. These calculations show that Li-doped graphite does not absorb hydrogen spontaneously. It was further found that the maximum storage of hydrogen molecules does not exceed a few percent. In addition, the diffusion barriers for hydrogen molecules to enter carbon nanotubes from the sidewalls – both with and without topological defects – was also considered. In all cases, relatively large diffusion barriers were found, indicating that it is unlikely that the mechanical processing of nanotubes would lead to an increased hydrogen storage.
Bibliography Type:	Description: (Last Updated: 09/10/2021)
Articles in Peer-reviewed Journals	Nakhmanson SM, Nardelli MB, Bernholc J. "Ab initio studies of polarization and piezoelectricity in vinylidene fluoride and BN-based polymers." Phys Rev Lett. 2004 Mar 19;92(11):115504. PMID: 15089148 , Mar-2004
Articles in Peer-reviewed Journals	Bernholc J, Nakhmanson SM, Buongiorno Nardelli M, Meunier V. "Understanding and enhancing polarization in complex materials." Computing in Science and Engineering. 2004 Nov-Dec;6(6):12-21. http://dx.doi.org/10.1109/MCSE.2004.78 , Dec-2004
Awards	Bernholc J. "Jesse Beams Award for Outstanding Research, American Physical Society, November 2003." Nov-2003
Papers from Meeting Proceedings	Meunier V, Buongiorno Nardelli M, Shelton W, Roland C, Bernholc J, Zacharia T. "Field Emission Properties of BN/C and BN@C hybrids." Three-Dimensional Nanoengineered Assemblies, MRS Fall Meeting, 2002. MRS Proceedings. Volume 739, Fall 2003. , Oct-2003
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